Technical Report

IX. Extension to GitTables

To evaluate the generalization of our contrastive framework beyond WikiDBGraph, we extend our analysis to GitTables [23], a corpus of heterogeneous web tables lacking explicit schema mappings. To adapt to this unsupervised setting, we introduce a self-supervised *table partition matching* task.

a) Task Formulation: Unlike WikiDBGraph, which utilizes shared Topic IDs for supervision, we generate ground truth via synthetic partitioning. We construct training triplets (x_a, x_p, x_n) where the anchor x_a and positive x_p are disjoint partitions of the same table T, while x_n is a partition from a randomly sampled table $T_j (j \neq i)$. We employ two partitioning strategies to simulate collaborative learning scenarios: (1) *Vertical Partitioning*, which splits columns to mimic feature-partitioned silos; and (2) *Horizontal Partitioning*, which splits rows to mimic sample-partitioned silos.

b) Experimental Setup: We use the partition `whole_tables_licensed` of GitTables [23], applying strict filtering criteria (minimum 50 rows, minimum 6 columns) to ensure sufficient content for partitioning. This results in 46,325 valid tables, from which we generate 72,685 triplets. The data is partitioned into an 80/10/10 split.

Similar to Section IV-B, input tables are serialized using both schema and sampled data: `[col_name]: val1, val2, ...`, with a sample size of 3 values per column and a maximum context of 50 columns. We fine-tune the `BGE-M3` model using InfoNCE loss objective with cosine similarity according to Eq. 4. Training is conducted for 10 epochs using the AdamW optimizer with a learning rate of 1×10^{-5} and a batch size of 32. During training, we sample 2 negatives per anchor.

TABLE XIV: GitTables relation prediction performance: Original (BGE-M3) vs contrastive fine-tuned like WikiDBGraph. Best in **bold**.

Metric	Original	Contrastive Learning
Accuracy	0.7454±0.0009	**0.9712±0.0010**
Precision	0.7666±0.0115	**0.9533±0.0014**
Recall	0.7067±0.0195	**0.9909±0.0006**
F1	0.7351±0.0056	**0.9718±0.0009**
AUC-ROC	0.8297±0.0008	**0.9888±0.0006**

c) Results: Evaluation is performed on the held-out test set (7,269 triplets) using a 1:1 balanced protocol across 5 random seeds. The results are detailed in Table XIV. Contrastive fine-tuning yields substantial improvements across all metrics compared to the pretrained baseline. Specifically, Accuracy increases from 74.54% to 97.12%, and the F1-score jumps from 0.7351 to 0.9718. The near-perfect AUC-ROC (0.9888) and Recall (0.9909) confirm that our objective effectively captures table provenance and semantic coherence even without explicit supervision. The result concludes that our approach can also generalize to other table corpora.

These results confirm that our contrastive objective generalizes beyond WikiDBGraph and captures table- or database-level provenance across diverse corpora. Crucially, the framework remains effective even when explicit supervision signals such as TIDs are unavailable, supporting its applicability to broader collections of tables and databases.

X. Performance Relation to Database Size

To investigate the robustness of collaborative learning across varying data scales, we analyze the performance improvement ($F1_{\text{FedAvg}} - F1_{\text{Solo}}$) relative to the local dataset size in Fig. 10. This analysis reveals two critical findings regarding the relationship between data volume and graph utility.

First, when employing naive string-based alignment (blue trend), we observe a distinct transition in graph utility relative to data scale: while small datasets ($< 10^2$ rows) benefit from external signals due to the "data starvation" effect, larger datasets ($> 10^3$ rows) suffer diminishing returns and potential negative transfer, as misaligned external features become noise that destabilizes already robust local models.

Second, employing semantic-aware alignment (DeepJoin [62], orange trend) effectively resolves this regression risk. The DeepJoin trend line remains consistently positive even for the top quartile of dataset sizes, with 77.4% of tasks exhibiting neutral or positive transfer. This confirms that the graph structure remains valuable for larger databases provided that preprocessing effectively resolves semantic correspondence, thereby attributing the regression risk to alignment quality rather than the intrinsic quality of the database connections.

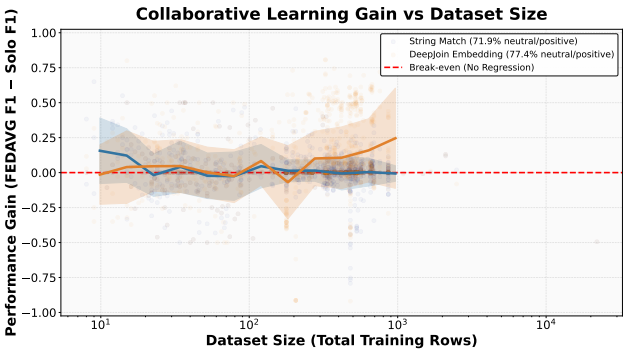

Fig. 10: Impact of dataset size on collaborative learning gain. The trend line represents the mean performance with variance indicated by the shaded region. Individual dots correspond to the performance of distinct database pairs.

XI. Discussion

This section summarizes key findings and implications for practical CL tasks derived from WikiDBGraph; a concise overview appears in Table XV. Some findings are consistent with established studies [4], [28], others parallel results reported in adjacent research areas [53], [54] but not previously for CL, and many are novel to our analysis. Collectively, these

TABLE XV: Summary of key findings and implications for practical Collaborative Learning (CL)

Finding / Implication for Collaborative Learning	Supporting Evidence (This Work)	Reference(s)
CL overall outperforms *Solo* training on real-world databases.	CL algorithms such as *FedAvg*, *FedProx*, and *FedOV* show improvements over *Solo* in both the overall analysis (Fig. 9) and the case studies (Tables IX, XI).	Aligns with [4], [28]
Advanced CL algorithms do not consistently outperform *FedAvg* on non-IID data.	Relational schemas typically contain non-IID data, producing performance discrepancies across clients, as observed in the overall analysis (Figs. 9) and the case study (Table IX).	Aligns with [4]
Databases in CL form highly interconnected networks with a long-tailed degree distribution; links should neither be ignored nor treated uniformly.	WikiDBGraph exhibits a long-tailed node-degree distribution (Fig. 4a), with low median but high maximum degrees (Table III).	Similar to social/web networks [53], [54]; newly reported for CL
New CL paradigms are needed for relational databases to avoid computationally infeasible full-table joins.	The "AllJoinSize" is infeasible on average (9.5×10^{16} rows; Table IV), making single-table approaches impractical.	Novel finding
CL algorithms should handle **hybrid** cases with partial instance and feature overlap, beyond ideal alignment assumptions.	Partial data and schema overlap are evidenced by the low `OverlapRatio` and `JaccardColumn` (Table V) and highlighted in the hybrid case study (Section VII-D).	Novel finding
Schema matching and entity resolution are critical, unresolved bottlenecks for real-world CL.	Manual schema/feature alignment in case studies (Section VII) yields strong *Combined* performance, whereas in automated evaluation only a subset of CL tasks benefit and average improvements are small for both *Combined* and CL algorithms.	Novel finding

results highlight the unique complexities of real-world data collaboration and motivate further investigation.

a) Effectiveness of Collaborative Learning: Both the overall analysis (Fig. 9) and the case studies (Tables IX, XI) show that CL algorithms generally outperform *Solo* training on real-world databases. However, advanced CL methods such as *FedProx* [29] do not consistently improve over simpler methods like *FedAvg* [28] on non-IID data. Both observations align with prior CL benchmarks [4], supporting WikiDBGraph as a valid benchmark for evaluating CL effectiveness.

b) Interconnection of Collaborative Learning: Databases in WikiDBGraph form highly interconnected networks with a long-tailed degree distribution. As evidenced by Fig. 4a and Table III, some nodes have high degree while many have low degree. Thus, most existing CL approaches [30], [55] that ignore links or treat all nodes uniformly are suboptimal. This property mirrors patterns in other real-world networks (e.g., social and web graphs) [53], [54] and underscores the need for graph-aware CL frameworks.

c) Infeasibility of Full Table Joins.: The average `AllJoinSize`—the estimated row count from joining all tables within a database—is 9×10^{16} (Table IV), rendering single-table approaches computationally infeasible. Consequently, CL methods that require full table joins are impractical for large real-world databases. Although such databases may be less common, they are often highly valuable due to their scale, calling for CL frameworks that operate directly on relational structures rather than a single prejoined table.

d) Hybrid Alignment in Collaborative Learning: Databases in WikiDBGraph are neither purely horizontally nor purely vertically aligned; instead, they exhibit *hybrid* alignment. This is supported by low `OverlapRatio` and `JaccardColumn` values in Table V and is further elaborated in the hybrid case study (Section VII-D). As a result, CL approaches that assume ideal horizontal or vertical alignment cannot handle many real-world cases, motivating methods that explicitly accommodate hybrid alignment.

XII. Future Directions

This section presents three main opportunities for data management research on WikiDBGraph.

Schema Matching: Analyses on WikiDBGraph indicate strong potential for schema matching [64], as identifying correct feature correspondences is crucial for learning accuracy. The dataset also provides ground truth for schema matching not only between two databases but also across multiple databases with complex relationships. WikiDBGraph surpasses prior datasets such as GitSchema [42] in scale and offers rich data beyond schemas alone. Its data distributions can further aid column alignment, enabling data-aware schema matching approaches.

Record Linkage and Data Mining: The prevalence of unaligned data suggests that record linkage [65] (a.k.a. entity alignment [66]) is inherently noisy [67]. Improving alignment in isolation, without feedback from downstream data mining, hinders task-aware alignment. A critical direction is to mine data under inaccurate or incomplete row alignment, which typically calls for a coupled data-mining framework [68].

Graph Mining: To incorporate databases in CL that cannot be directly joined, graph mining [69] techniques must be extended to support CL over database graphs—tables connected by foreign keys. This join challenge calls for new graph-based CL methods beyond current table-centric approaches.